

Rajasthan Board Class 12

Subject – Chemistry

Time: $3\frac{1}{4}$ Hours

M.M: 56

Section - A

1. Write the value of axial distances and axial angles of triclinic crystal.

Answer:

Let a, b and c be three sides of a unit cell; α , β , γ be the angles between b and c, a and c, a and b respectively. Then, for triclinic crystal, $a \neq b \neq c$ and $\alpha \neq \beta \neq \gamma \neq 90^\circ$.

2. Write the definition of Kohlrausch law of independent migration of ions.

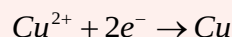
Answer:

The law states that limiting molar conductivity of an electrolyte can be represented as the sum of the individual contributions of the anion and cation of the electrolyte.

3. Give half-cell equation of Daniell cell takes place at cathode.

Answer:

The half-cell equation of Daniell cell that takes place at cathode is represented as given below:



4. Write the unit of velocity constant for second order reaction.

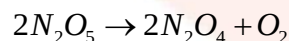
Answer:

Unit of velocity constant for second order reaction is $\text{L mol}^{-1} \text{s}^{-1}$.

5. For reaction $2\text{N}_2\text{O}_5 \rightarrow 2\text{N}_2\text{O}_4 + \text{O}_2$ the half life time is 6.93 sec, determine the rate constant.

Answer:

We have,



We know that the rate of reaction is related to half-life period.

$$k = \frac{0.693}{t_{\frac{1}{2}}}$$

$$\Rightarrow \frac{0.693}{6.93} \left[\because t_{\frac{1}{2}} = 6.93 \right]$$

$$\Rightarrow k = 0.1$$

Hence, the rate constant is 10s^{-1} .

6. Transition element form interstitial compound. Give one reason.

Answer:

It is because the large sizes of transition elements, they have lots of interstitial sites and can trap smaller elements such as H, C, and N etc.

7. Write the formula of each alkyl alcohol and benzyl alcohol.

Answer:

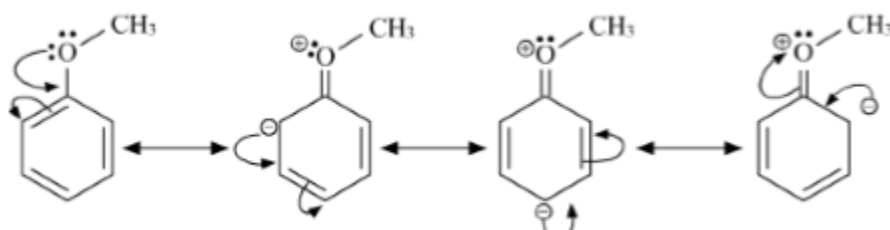
Alkyl alcohol: $R-O-H$; where R is an alkyl group.

Benzyl alcohol: $C_6H_5-CH_2-OH$

8. Give resonating structures of anisole.

Answer:

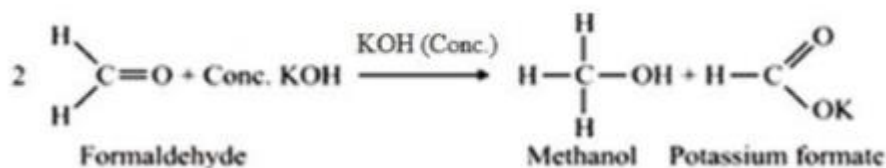
Resonating structure of anisole,



9. Write chemical equation of Cannizzaro reaction.

Answer:

Aldehydes which do not have any α - hydrogen atom, undergo self oxidation and reduction.



10. Chloroacetic acid is strongest acid than acetic acid give one reason.

Answer:

Inductive effect due to the presence of chlorine group increases the polarity of hydroxide bond in chloroacetic acid which makes it more acidic as compared to acetic acid.

11. Write the name of following compounds:-

(A) $C_6H_5N_3^+HSO_4^-$



Answer:

(A) Benzene diazonium hydrogen sulphate

(B) p-hydroxyazobenzene

12. . Identify the type of following drugs and write its types.

(A) Valium (B) Cimetidine

Answer:

(A) Valium → Tranquillizer

(B) Cimetidine → Antacid

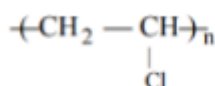
13. Write the chemical formula of teflon and P.V.C.

Answer:

Chemical formula of Teflon is:



And the chemical formula of Teflon P.V.C are



Section - B

14. (A) Give any one difference between anisotropy and isotropy nature of solid.

(B) Electric conductivities of solid (X) is $10^4 - 10^7 \text{ ohm}^{-1}\text{m}^{-1}$ and solid

(Y) is $10^{-20} - 10^{-10} \text{ ohm}^{-1}\text{m}^{-1}$. Identify solid (X) and the solid (Y) and write their name.

Answer:

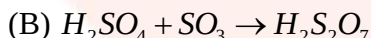
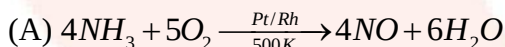
(A) The difference between anisotropic and isotropic are in anisotropy, the properties of a material vary with different directions. While, in isotropy, the properties of a material are same in all directions

15. Write only chemical equations for following reactions.

(A) When Ammonia react with atmospheric oxygen in the presence of catalyst.

(B) When sulphuric acid react with sulphur trioxide.

Answer:



16. (A) Write only the types of hybridisation of central atom present in XeF_2 and XeF_4 .

(B) Explain the reason of different length of equatorial and axial bond in the structure of PCl_5 .

Answer:

(A) XeF_2 : Hybridisation of central atom is sp^3d

XeF_4 : Hybridisation of central atom is sp^3d^2

(B) There are three equatorial bonds and two axial bonds in PCl_5 . And the axial bonds are longer than equatorial bonds because of greater repulsion from equatorial bonds. The

2-axial bonds are at 90° degree angle to 3-equatorial bonds while all equatorial bonds are at 120° angle to each other.

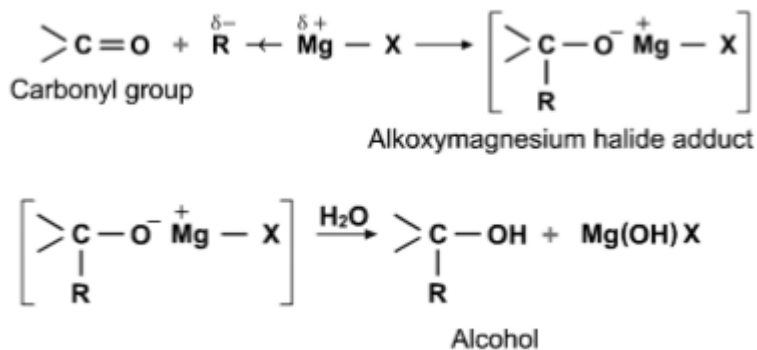
17. (A) Explain denaturation of alcohol.

(B) Give chemical equation to obtain primary alcohol from grignard reagent.

Answer:

(A) The process of making alcohol unfit for human consumption, by adding small quantity of CuSO_4 and pyridine.

(B) Chemical equations are as follows:



18. (A) Give two examples of biodegradable polymer.

(B) Give the name of both monomers used in formation of terylene.

Answer:

(A) Two examples of biodegradable polymers are polyglycolic (PGA), Polyhydroxy butyrate (PGB).

(B) Monomers of terylene are Terephthalic acid (1, 4 benzene dicarboxylic acid) and ethylene glycol (1, 2 ethanediol).

19. (A) Write any two differences between soap and detergent.

(B) In which work saccharine is used.

Answer:

(A) The difference between soap and detergent are as given below:

Soap	Detergent
1. They are sodium or potassium salts of fatty acids	1. They are sodium or potassium salts of sulphonic acids.
2. They have $-\text{COONa}$ group.	2. They have $-\text{SO}_3\text{Na}$ group

3. They do not work well with hard water, acidic water and saline water.

3. They work well with hard water, acidic water and saline water.

(B) Saccharin is an artificial sweetener and is used to sweeten products such as drinks.

20. (A) Write the name of two metals which used in maximum composition in mischmetal.

(B) Calculate the value of magnetic moment of V^{+2}

Answer:

(A) Major constituents of mischmetal are La and Ce

(B) Electronic configuration $V^{+2} = [Ar] 4s^0 3d^3$

Since, it has 3 unpaired electrons

$$\text{Magnetic moment } (\mu) = \sqrt{n(n+2)} BM$$

Where, n is the number of unpaired electrons.

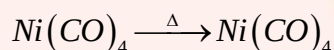
$$\text{Therefore, } V^{+2} = \sqrt{3(3+2)} \approx 3.9 BM$$

21. (A) Give both chemical equations of 'Mond process' used in nickel refining.

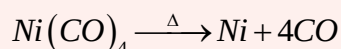
(B) Draw a neat and labelled diagram of zone refining process.

Answer:

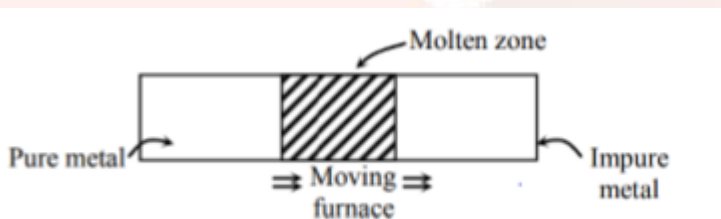
(A) The impure nickel is heated with excess carbon monoxide to form the gas nickel carbonyl, leaving the impurities as solids like:



And the mixture of excess carbon monoxide and nickel carbonyl is heated again where nickel tetracarbonyl decomposes to give pure nickel.



(B)



OR

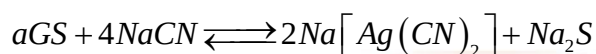
21. (A) Give both chemical equations of leaching of silver in silver metallurgy by using sodium cyanide.

(B) Draw a neat and labelled diagram of electrolytic cell for the extraction of aluminum.

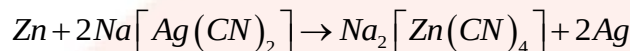
Answer.

(A) The process of leaching of silver by using sodium cyanide is given below:

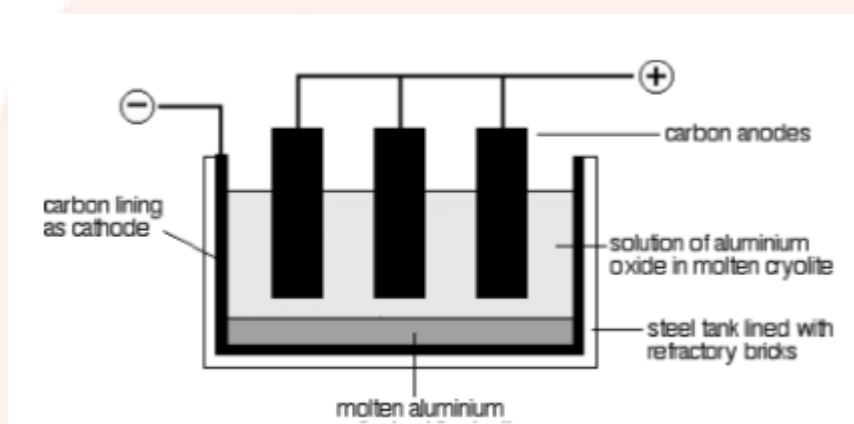
Silver is extracted from the ore-argentite Ag_2S . The ore is crushed, concentrated and then treated with sodium cyanide solution. This reaction forms sodium argent cyanide $(Na[Ag(CN)_2])$



And the solution of sodium argento cyanide combines with zinc dust to form precipitated silver and sodium cyanozincate.



(B)



22.

(A) Write the name of metal used in sacrificial electrode for the prevention of corrosion of iron metal. How it prevents the corrosion?

(B) If Λ_M^0 for $NaCl$, HCl and CH_3COOH are 110, 100 and $390 \text{ Scm}^2 \text{ mole}^{-1}$ respectively. Determine the value of CH_3COONa .

Answer:

Zinc is used in sacrificial electrode for the prevention of corrosion of iron metal. Because sacrificial anode is a very reactive metal object or coating whose purpose is to corrode and dissolve, sacrificing itself, while a protected metal object remains intact.

According to the Kohlrausch law we get that,

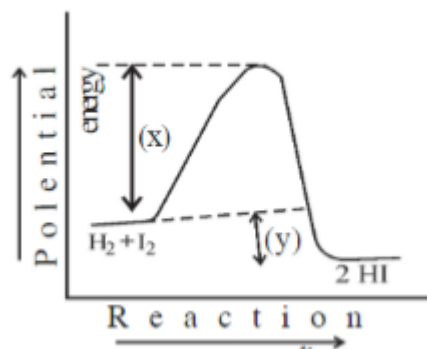
$$\Lambda_{CH_3COOH} = \Lambda_{CH_3COO^-} + \Lambda_{H^+}$$

$$\Rightarrow \Lambda_{CH_3COOH} = \Lambda_{CH_3COONa} + \Lambda_{HCl} - \Lambda_{NaCl}$$

$$\Rightarrow 390 = \Lambda_{CH_3COONa} + 100 - 110$$

$$\Rightarrow \Lambda_{CH_3COONa} = 400 \text{ Scm}^2 \text{ mol}^{-1}$$

23.



(A) Write suitable name of (X) and (Y) denoted in above graph.

(B) Give definition of only (X) denoted in graph.

Answer:

(A) X $\rightarrow$ Activation energy (E_a)

Y $\rightarrow$ Enthalpy of reaction

(B) The extra quantity of energy which must be possessed by reactant molecules so that collision between reactant molecules is effective and leads to the formation of product molecules is known as activation energy.

24. (A) Write the name of diseases due to deficiency of vitamins 'A' and vitamins 'B'

(B) On the basis of 'Sugar differentiate D.N.A and R.N.A.

Answer:

(A) Vitamin A $\rightarrow$ Night blindness

Vitamin B $\rightarrow$ Beri berry

(B) DNA consist of β -D- oxyribose sugar.

RNA consist of β -D- ribose sugar

Section – C

25. (A) Due to low concentration of oxygen in blood, climber become weak and unable to think clear-

(i) Write specific name of above condition. (ii) Explain the reason of such condition.

(B) 30 gm of ethanoic acid present in 100gm of water, determine molality of ethanoic acid in water.

Answer:

(A)(i) Anoxia

(ii) At high altitudes the partial pressure of oxygen is less than that at the ground level. Which leads to low concentration of oxygen in the blood and tissues.

(B) We have,

Mass of ethanoic acid = 30gm

Mass of water = 100gm

Molar mass of CH_3COOH



$$= 24 + 4 + 32$$

$$= 60\text{gm}$$

Therefore, the molality,

$$(m) = \frac{\text{Weight of solute} \times 100}{\text{Molar weight of solute} \times \text{Weight of solvent}}$$
$$= \frac{30 \times 100}{60 \times 100}$$
$$= 5$$

Hence, molality of ethanoic acid in water is 5 mol/kg.

26. Compound [A] is an aromatic amine which reacts with $\text{NaNO}_2 + \text{HCl}$ at 273-278K and forms compound [B]. Compound [B] react with HBF_4 and the obtained product on further heating, in the presence of NaNO_2 and Cu form compound [C]. Compound [C] reduced in the presence of $\text{Sn} + \text{HCl}$ to re-formed compound [A]. Write general name of 'A', 'B' and 'C' and write equation of all reactions involved.

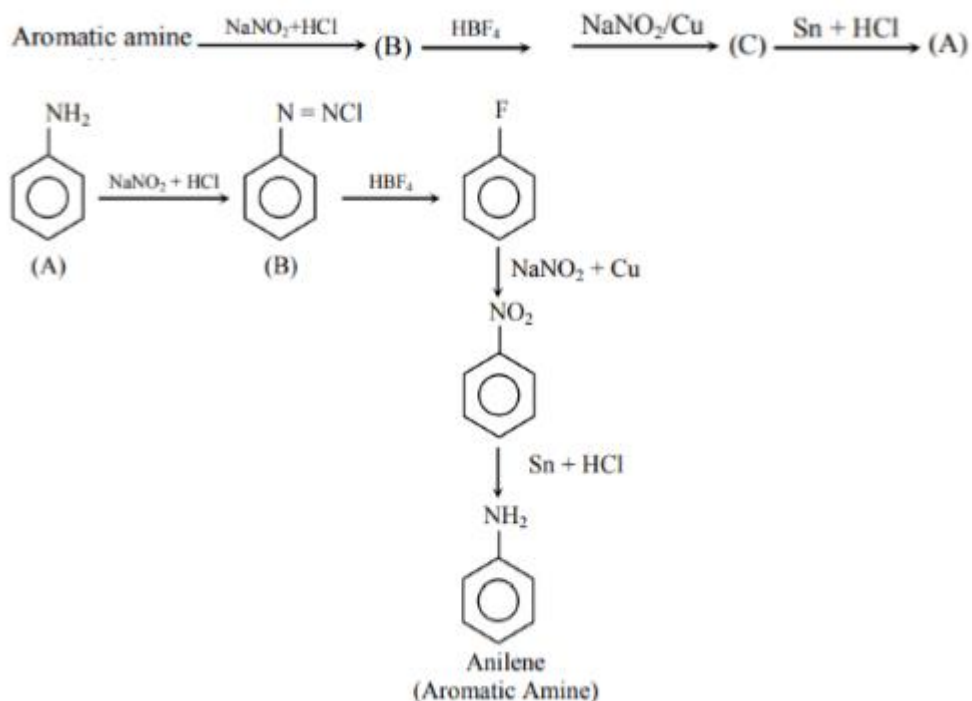
Answer:

We have,

Compound [A] = Aniline

Compound [B] = Benzene diazonium chloride

Compound [C] = Nitro Benzene



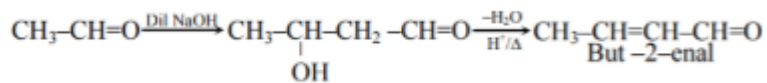
27. Write the equation of chemical reaction and name of reaction to obtain following compounds (A) and (B) from CH_3CHO .

(A) But – 2 –enal

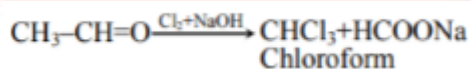
(B) Chloroform

Answer:

(A) Aldol condensation reaction:



(B) Chloroform reaction:



OR

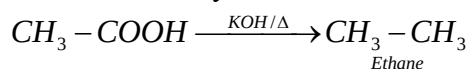
27. Write the equation of chemical reaction and name of to obtain following compounds (A) and (B) from CH_3COOH .

(A) Ethane

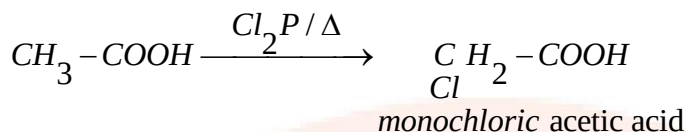
(B) Mono chloro acetic acid.

Answer:

(A) Kolbe's electrolysis



(B) Hell-volhard zelinsky Reaction (H.V.Z reaction):



Section – D

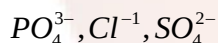
28. 1. One of the factory of city was banned by pollution board because factory owner for not arranging a chamber on chimney.

(a) Write the name of chamber which was not installed by factory owner.

(b) Draw neat and labelled diagram of above chamber.

2. (a) Define coagulating value of sol.

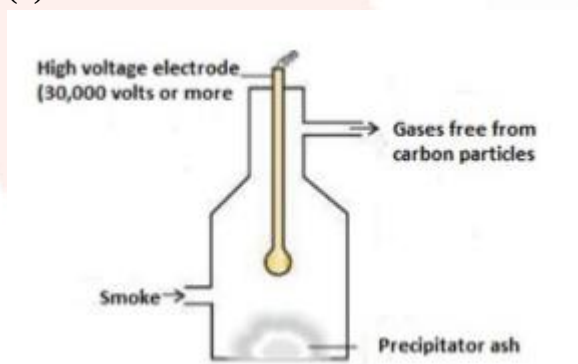
(b) Arrange the following ions in increasing order on the basis of their coagulating values.



Answer:

1. (a) Cottrell's smoke precipitator

(b)



2. (a) Coagulating values is termed as the minimum concentration of an electrolyte in millimoles per liter required to bring about the complete coagulation of a sol.

(b) $PO_4^{3-}, Cl^{-1}, SO_4^{2-}$ if the value of coagulation is higher than the coagulating power of an ion.

OR

28. 1. (a) Which type of emulsion is 'Vanishing cream'. Write its appropriate name.

(b) Draw neat and labeled diagram of above emulsion.

2. (a) What is shape selective catalysis?

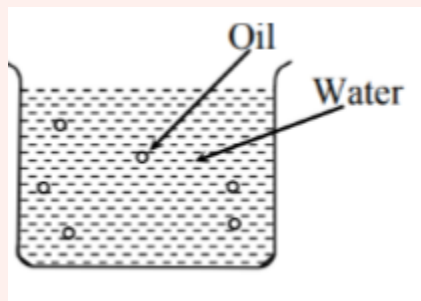
(b) Write the name of adsorbent used in following –

(i) To remove colouring matter from solution. (ii) In gas mask.

Answer.

1. (a) Vanishing cream is an emulsion with oil in dispersed phase and water in dispersion medium.

(b)



2. (a) Shape selective catalysis is defined as the catalytic reaction which depends on the pore size of the catalyst and also on the shape and size of the reactant and product molecule.

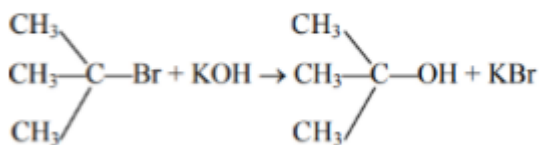
(b) (i) Animal Charcoal.

(ii) Activated charcoal or mixture of adsorbents.

29. 1. Write the chemical equation of following reaction. (a) Swarts reaction (b) Sandeyer's reaction.

2. (a) (i) $\text{CH}_3 - \text{Br} + \text{KOH} \rightarrow \text{CH}_3 - \text{OH} + \text{KBr}$

(ii)

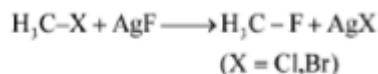


Which types of nucleophilic substitutions are in above reaction (i) and (ii)

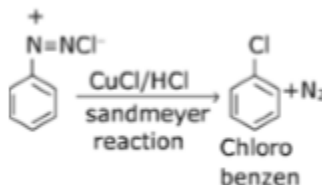
(b) Write two differences between mechanism of reaction (i) and (ii)

Answer:

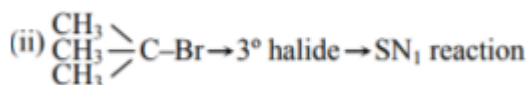
Swartz reaction is defined as alkyl chloride is heated in the presence of a metallic fluoride like AgF , Hg_2F_3 , SbF_3 or CoF_2 .



Sandmeyer's reaction is a reaction in which aryl diazonium salt is treated with Cu(I) .



2. (a) (i) $\text{CH}_3-\text{Br} \rightarrow 1^\circ \text{halide} \rightarrow \text{SN}_2 \text{ reaction}$



(b)

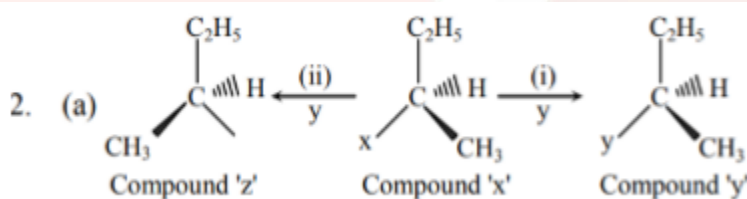
SN_2 mechanism	SN_1 mechanism
(i) No intermediate formed only transition state is formed	(i) Intermediate is carbocation.
(ii) It is a 2nd order reaction.	(ii) It is a 1st order reaction.

OR

29. 1. Write the chemical equation of following reaction

(a) Wurtz reaction

(b) Finkelstein reaction

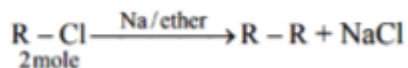


Write the names of process to obtain compound (y) and compound (z) from compound (x) in above nucleophilic reaction (i) and (ii).

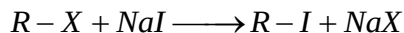
(b) Which type of nucleophilic substitution mechanism involved to obtain compound (z) from compound (x). Write the name of mechanism.

Answer:

- (a) Wurtz reaction is defined as the reaction in which alkyl halides on treatment with sodium metal in dry ethereal (free from moisture) solution give higher alkanes.



(b) Finkelstein reaction is defined as the alkyl iodides are often prepared by the reaction of alkyl chlorides/ bromides with NaI in dry acetone.



2. (a) (i) Y is formed by retention process
(ii) Z is formed by inversion process
(b) It is bimolecular nucleophilic substitution reaction and the name of mechanism is $\text{S}_\text{N}2$.

30. (i) Define primary and secondary valency of metal ions proposed by Werner theory.

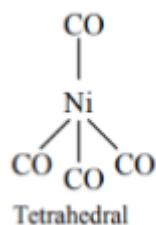
(ii) Write primary and secondary valency of Co in $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$.

(iii) Write the structural formula of $\text{Ni}(\text{CO})_4$ and $\text{Cr}(\text{CO})_6$.

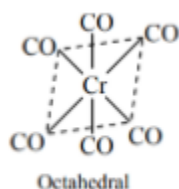
(iv) Draw the structure which shows Synergic bonding interaction in a carbonyl Complex.

Answer:

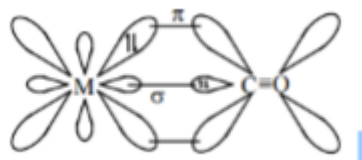
- (i) Werner's theory states that metal ions have two types of valency primary and secondary in which,
- Primary valency is said to be satisfied by negative ions only, and depends on the oxidation state of central metal.
 - Secondary valency can be satisfied by positive ligand, negative or neutral molecule and it equals to the co-ordination number of central metal.
- (ii) Co in $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
Primary valency $\text{Co} = +3$
Secondary valency $\text{Co} = 6$
- (iii) Structural formula of $\text{Ni}(\text{CO})_4$ and,



Structural formula of $\text{Cr}(\text{CO})_6$



(iv) Structure showing synergic bonding interaction in a carbonyl complex:



OR

30. (i) Define linkage and ionization isomerism.

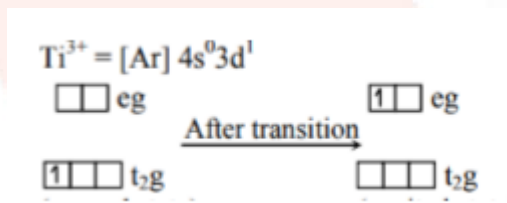
(ii) Which isomerism show by $[Co(NH_3)_6][Cr(CN)_6]$ $[Cr(NH_3)_6][Co(CN)_6]$. Write name.

(iii) Draw the diagram which show the transition of an electron in $[Ti(H_2O)_6]^{3+}$.

(iv) What impact takes place on metal complex due to transition of an electron in $[Ti(H_2O)_6]^{3+}$.

Answer:

- (i) Isomerism is shown by the coordination compounds having ambidentate ligands is known as Linkage isomerism. While, isomerism arises due to exchange of ionisable anion with anionic ligand is known as Ionization isomerism.
- (ii) $[Co(NH_3)_6][Cr(CN)_6]$ $[Cr(NH_3)_6][Co(CN)_6]$ show co-ordination isomerism.
- (iii) $[Ti(H_2O)_6]^{3+}$



- (iv) Transition of an electron in $[Ti(H_2O)_6]^{3+}$ imparts to the complex.